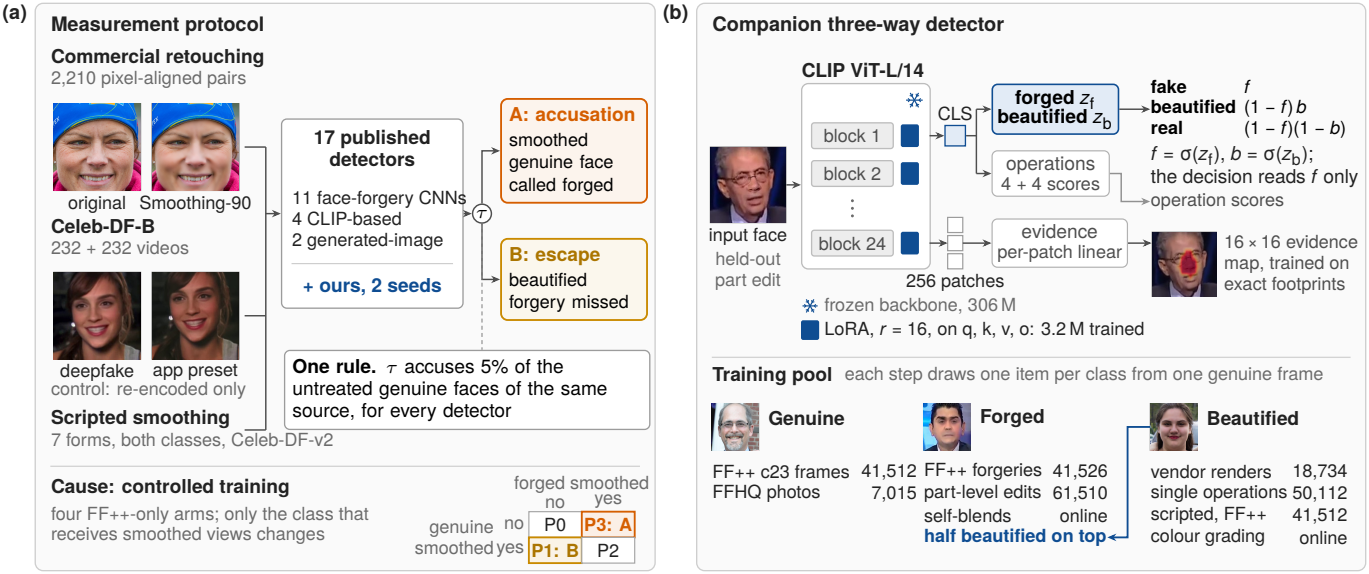

(a)
Measurement protocol
Commercial retouching
2,210 pixel-aligned pairs
original
Smoothing-90
Celeb-DF-B
232 + 232 videos
deepfake
app preset
control: re-encoded only
Scripted smoothing
7 forms, both classes, Celeb-DF-v2
17 published detectors
11 face-forgery CNNs
4 CLIP-based
2 generated-image
+ ours, 2 seeds
τ
A: accusation
smoothed genuine face called forged
B: escape
beautified forgery missed
One rule. τ accuses 5% of the untreated genuine faces of the same source, for every detector
Cause: controlled training
four FF++-only arms; only the class that receives smoothed views changes
forged
smoothed
no
yes
genuine
smoothed
P0
P3: A
P1: B
P2
(b)
Companion three-way detector
CLIP ViT-L/14
input face
held-out part edit
block 1
block 2
block 24
256 patches
CLS
forged z_f
beautified z_b
operations 4 + 4 scores
evidence per-patch linear
fake f
beautified (1 − f) b
real (1 − f)(1 − b)
f = σ(z_f), b = σ(z_b); the decision reads f only
operation scores
16 × 16 evidence map, trained on exact footprints
frozen backbone, 306 M
LoRA, r = 16, on q, k, v, o: 3.2 M trained
Training pool
each step draws one item per class from one genuine frame
Genuine
FF++ c23 frames 41,512
FFHQ photos 7,015
Forged
FF++ forgeries 41,526
part-level edits 61,510
self-blends online
half beautified on top
Beautified
vendor renders 18,734
single operations 50,112
scripted, FF++ 41,512
colour grading online